

Tusher Chandra Mondol

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Professional Summary

HPC systems researcher and Ph.D. candidate in Computer Science specializing in large-scale interconnection networks and system-level performance optimization. Research to date has focused on Dragonfly systems, including adaptive routing, process mapping, congestion analysis, and packet/MPI-level simulation across traditional HPC and AI workloads. Ongoing work extends to node allocation and inter-job interference, with broader interests in systems for AI and AI for systems toward efficient and scalable next-generation HPC platforms.

Education

Florida State University

Ph.D. Candidate in Computer Science (Advisor: Dr. Xin Yuan)

Tallahassee, FL

Expected Apr 2027

Research Interests: HPC systems, interconnection networks, systems for AI, and AI for systems

Khulna University of Engineering & Technology

B.Sc. in Computer Science and Engineering

Khulna, Bangladesh

Mar 2019

Work Experience

Graduate Research & Teaching Assistant (alternating appointments)

Aug 2024 – Present

Department of Computer Science, Florida State University

Tallahassee, FL

- Conduct research on HPC interconnection networks spanning adaptive routing, process mapping, node allocation, and inter-job interference, with large-scale performance evaluation using SST and BookSim.
- Led recitations, developed assessments, and provided student support across computer science courses including Concurrent, Parallel, and Distributed Programming; Parallel and Distributed Systems; Binary Formats and Linkers; and Computer and Network Administration.

Graduate Research Assistant

Aug 2021 – Jul 2024

Department of Educational Psychology and Learning Systems, Florida State University

Tallahassee, FL

- Designed immersive simulation environments for nursing/healthcare education with AI-driven virtual agents, adaptive dialogue, speech interaction, and Flask-based data services; contributed to usability studies and pilot evaluations.
- Developed a Bayesian-network-based competency assessment and level-recommendation system and an interactive Unity simulation environment for game-based mathematics education.

These projects contributed to three ICLS publications.

Software Engineer

Feb 2020 – Nov 2020

Nascenia

Dhaka, Bangladesh

- Developed backend automation systems using Ruby on Rails for data processing, system monitoring, and workflow management for Robi Axiata Limited, reducing manual intervention in operational data workflows.

Selected Publications & Research Output

- **T. C. Mondol**, X. Yuan, Z. Lan. “Constrained Multi-Agent Reinforcement Learning-Based Routing for Dragonfly Networks.” Under review, IEEE International Parallel & Distributed Processing Symposium (IPDPS), 2027.
- **T. C. Mondol**, X. Yuan. “DraCO-Map: Beyond Locality in Hierarchical Process Mapping for Dragonfly Networks.” Manuscript in preparation, 2026.
- R. S. Chaulagain, **T. C. Mondol**, S. Bhowmik, X. Yuan. “Exploiting Software-Defined Networking Technology for Improving UGAL Routing in Dragonfly Networks.” *25th IEEE International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, 2025.
- F. Ke, X. Yuan, N. Zhang, C. Kim, A. Barrett, R. Shin, **T. C. Mondol**, J. Whyte. “From Practice Partner to Co-Learning: Designing Role-Fluid AI Partnership for Skills Development.” *International Conference of the Learning Sciences (ICLS)*, 2026.
- C. Kim, F. Ke, **T. C. Mondol**, S. Bhowmik, J. Whyte, X. Yuan. “Leveraging AI-Powered Simulation for Transitional Care: A Pilot Study in Usability and Learning Design.” *International Conference of the Learning Sciences (ICLS)*, 2026.
- L. West, F. Ke, C. Dai, B. J. Kim, R. S. Chaulagain, **T. C. Mondol**, J. Xu, R. Kuba, S. Jung. “Effects of Level Trajectory on Mathematical Gameplay.” *International Conference of the Learning Sciences (ICLS)*, 2022.
- **T. C. Mondol**, H. Iqbal, M. M. A. Hashem. “Deep CNN-Based Ensemble CADx Model for Musculoskeletal Abnormality Detection from Radiographs.” *International Conference on Advances in Electrical Engineering (ICAEE)*, 2019.

Research Experience

Process Mapping for Dragonfly Networks

Manuscript

- Evaluated heuristic, metaheuristic, and graph-partitioning-based process mapping approaches for Dragonfly systems, identifying limitations in locality-oriented methods for load balancing, induced traffic, and physical placement.
- Developed DraCO-Map, a hierarchical graph-partitioning-based framework that jointly considers communication locality and network load while accounting for induced traffic and physical placement across Dragonfly groups and routers.
- Evaluated traditional HPC and AI workloads using ALCF job traces and representative Polaris and Aurora configurations, achieving an average application-performance improvement of X% over existing mapping approaches, together with improved locality and network-load distribution.

Adaptive Routing for Dragonfly Networks

CCGrid 2025; IPDPS 2027 (under review)

- Investigated software-defined approaches for improving UGAL routing in Dragonfly networks, contributing to a peer-reviewed publication at CCGrid 2025.
- Identified path-enumeration and learning-dynamics challenges in applying MARL to constrained routing; developed a general constraint-aware MARL routing framework and proposed throughput degradation factor (TDF) to quantify candidate-path enumeration quality.
- Instantiated the framework as UGAL-Q and evaluated it across HPC and deep-learning workloads, achieving up to 66.47% lower communication time than prior MARL-based Dragonfly routing schemes while approaching the globally informed UGAL-G baseline.

Fragmentation-Aware Node Allocation and Job Interference

Ongoing

- Investigating communication- and fragmentation-aware node allocation using group/router-level resource availability and workload characteristics to guide compact-versus-spread allocation decisions, with ongoing work extending to inter-job interference using production-style workload traces.

Large-Scale Interconnect Simulation and Performance Analysis

- Built and extended SST (Merlin, Ember, and Firefly) and BookSim workflows to study traditional HPC and AI workloads, MPI motifs, routing and mapping algorithms, topology variants, dynamic traffic, congestion, performance variability, and scalability.

Technical Skills

Programming	Python, C/C++, C#, Ruby
Parallel Computing	MPI, OpenMP, CUDA, NCCL, SIMD (SSE/AVX)
HPC Simulation	SST (Ember, Firefly, Merlin), BookSim
Machine Learning	PyTorch, TensorFlow, Scikit-learn
Tools & Frameworks	METIS, Git, L ^A T _E X, MySQL, Unity, Ruby on Rails, Flask

Selected Projects

- Investigated Dragonfly global-link arrangements using genetic-algorithm and randomized strategies; extensive evaluation showed limited performance sensitivity to alternative arrangements under the studied workloads and configurations.
- Explored machine-learning-based estimation of the routing bias used by UGAL-L to adapt routing decisions to observed network conditions rather than relying on a fixed bias parameter.
- Implemented a neural network from scratch in C/C++ for MNIST and accelerated execution using loop optimization, SSE/AVX vectorization, OpenMP, MPI, and CUDA.

Leadership & Honors

- Chevron International REACH Scholarship
- Champion, KUET University Day Project Showcasing (2017)
- Champion, KUET CSE Intra-department Idea Contest (2017)
- President, KUET CSE Association Student Body (2017–2019)
- Vice-President, Special Group Interested in Programming Contest, KUET (2017–2019)